

Structure of the message ID

The message ID consists of:

- System (D0 ...) or axis ID (D1...)
- 1- or 3-part diagnostic number, derived from the main error group, error sub-group and error number

→ Main/sub-groups of the message ID

- Instance for identifying the affected component

Message ID						Description
[1]	[2]	[3]	[4]	[5]	[6]	System message "Over-current monitoring"
D	0.	16842763			0	1-part diagnostic number
D	0.	01	01	00011.	0	3-part diagnostic number – Main group = 01 → current – Sub-group = 01 → I ² t – Error number = 00011 → overcurrent error

Tab. 729: Example of message ID Over-current monitoring

D	System/axis	ID diagnostic number			Instance
		Main group	Subgroup	Number	
.	.	.	.	.	[6] → Identifier of the affected component with similar components. Example: Encoder 1 and encoder 2.
.	.	.	.	.	
.	.	.	.	.	
.	.	.	.	.	
.	.	.	.	.	[5] → ID error number
.	.	.	.	.	
.	.	.	[4] →	Sub-group as a function of the main group, e.g. logic supply, DC link supply	
.	.	.	.	[3] → Main group of the error, e.g. current, voltage, temperature error	
.	.	.	.		
.	[2] →	Identification of the system or axis: 0 = system x = number of the related axis (1 = axis 1, 2 = axis 2, ...)			
[1] →	Identification D = Diagnostics				

Tab. 730: Message ID Dx ...

Representation of the diagnostic number

The diagnostic number is optionally displayed in 2 variants:

- 1-part = 1 resulting number from main group, sub-group and error number
- 3 parts = 3 individual numbers for main group, sub-group and error number.

This representation is used e.g. in the plug-in.

The variants are determined via the hexadecimal/decimal conversion. Proceed as follows to determine the 1-part diagnostic number:

- Combine main group, sub-group and error number byte by byte as a hexadecimal 4-byte value:
Main group = byte 4
Sub-group = byte 3
Error number = byte 2 and 1
- Convert the resulting hexadecimal 4-byte number into a decimal number.
This results in the 1-part diagnostic number.

Main group	Subgroup	Error number	Comment
Byte 4	Byte 3	Byte 2 ... 1	
02	02	00031	3-part diagnostic number, decimal
Conversion of dec → hex			

Main group	Subgroup	Error number	Comment
0x 02	0x 02	0x 00 1F	Individual values, hexadecimal
Grouping			
0x 02 02 00 1F			Grouping of the individual values (4-byte number), hexadecimal
Conversion hex → dec			
33685535			1-part diagnostic number, decimal

Tab. 731: Display variants of the diagnostic number (example)



If the device profile only allows 16 bit, only the ID error number **5** is transferred → Tab. 729 Example of message ID Over-current monitoring.

The 1-part diagnostic number of the currently most serious diagnostic event can be read out via the following parameters:

Type of error	Parameters	CIA object	PNU
System-related errors	Instance 0: Px.315.0.0	0x2145.0B ... 0C	3376.0
System-related or axis-related errors	Instance 1: Px.315.1.0	0x2145.0C	3377.0

Tab. 732: Actually most serious error

Main/sub-groups of the message ID

Main group		Subgroup	
01	Current	01	Short circuit
		02	I _{pt}
		03	Braking resistor
02	Voltage	01	Supply
		02	DC link circuit
		03	Principal voltage
		04	Encoder supply
03	Temperature	01	Device
		02	Output stage
		03	Motor
05	Motion	01	Homing
		02	Motion control
		03	Interpolation
06	Configuration/parameterization	00	No allocation
		02	Critical limits
		05	Parameter set
07	Monitoring	01	Limitations
		02	Motion monitoring
		03	Critical limits
		04	Zero angle detection
		05	Analogue input
		11	Friction
08	Communication	00	No allocation
		03	PROFINET
		04	EtherCAT®
		06	EtherNet
		09	PROFIdrive

Main group		Subgroup	
08	Communication	12	CiA 402
		13	EtherNet/IP
		14	MP
09	Safety engineering	00	No allocation
		01	STO
		02	SBC
10	Internal hardware	01	Module error
11	Software	00	No allocation
		01	Exception
		02	Task
		03	File system
		04	Firmware update
		05	Device configuration
		06	LibRTE
		07	Warm start
		08	Version management
12	Maintenance	01	Operating time
13	Various	01	Diagnostics
		02	Auto-tuning
16	External device	01	CDSB
17	Security (data)	01	User login
18	Encoder	00	No allocation
		01	EnDat
		02	Hiperface
		03	Quadrature (incremental encoder)
		04	Nikon A
		05	BiSS C
		06	Sin/Cos
		07	ProfiDrive

Tab. 733: Main/sub-groups of the message ID

9.4.3 Message directory

All messages are stored in the volatile message directory of the device. The sequence of the entries is determined by the time when they occurred and the severity. The message directory is sorted as follows:

- Severity in descending order (messages with a higher degree of severity come first)
- Time stamp in ascending order (older messages come first)

The plug-in in the 'Diagnosis' context offers access to the message directory, 'Device State' diagnostics panel.

The CDSB displays active messages: 'Diagnosis' menu, 'Active Messages' command.